

Name: Benedicto, Ruel Jr A.	Date Performed: 12/09/2025
Course/Section: CPE212 - CPE31S1	Date Submitted: 12/09/2025
Instructor: Engr. Robin Valenzuela	Semester and SY: 2nd Semester S.Y. 2025 - 2026

Activity 1: Configure Network using Virtual Machines

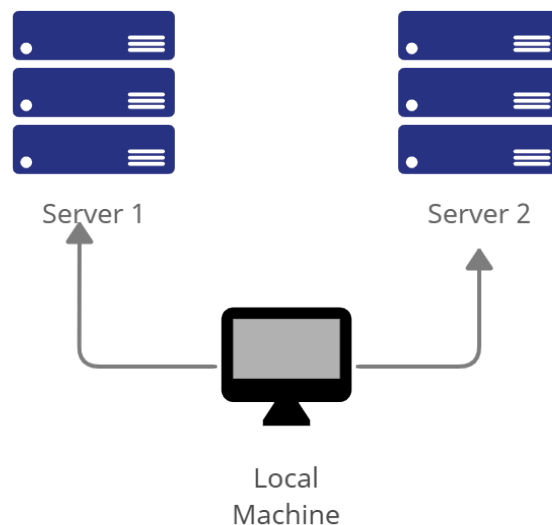
1. Objectives:

- 1.1. Create and configure Virtual Machines in Microsoft Azure or VirtualBox
- 1.2. Set-up a Virtual Network and Test Connectivity of VMs

2. Discussion:

Network Topology:

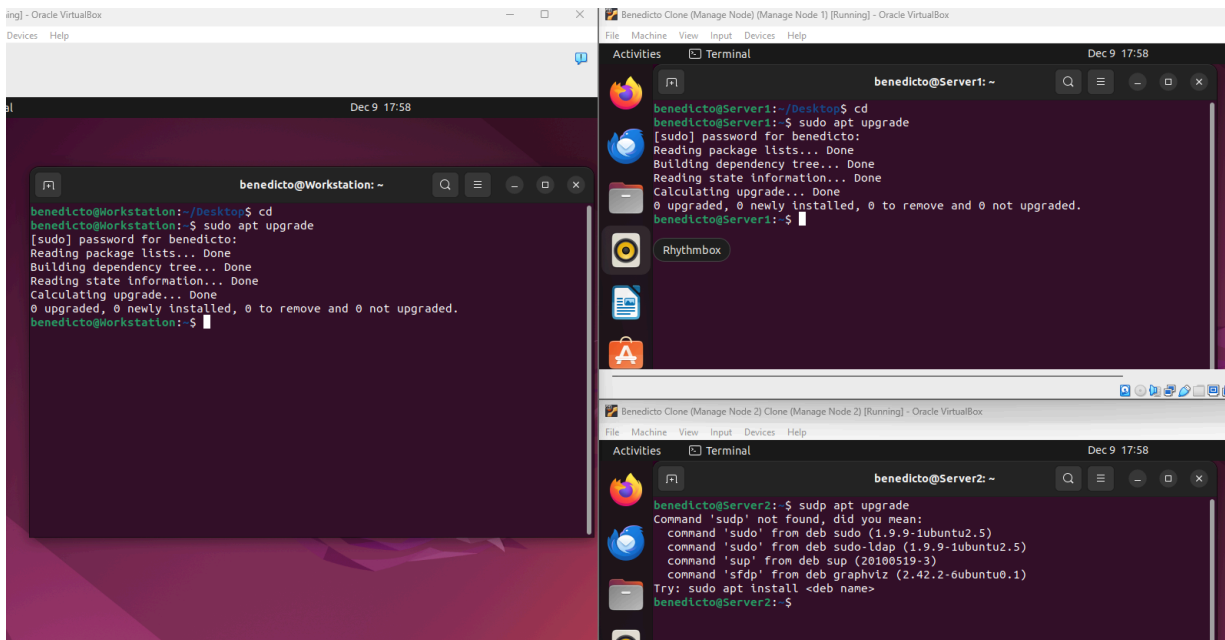
Assume that you have created the following network topology in Virtual Machines, *provide screenshots for each task*. (Note: it is assumed that you have the prior knowledge of cloning and creating snapshots in a virtual machine).



Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

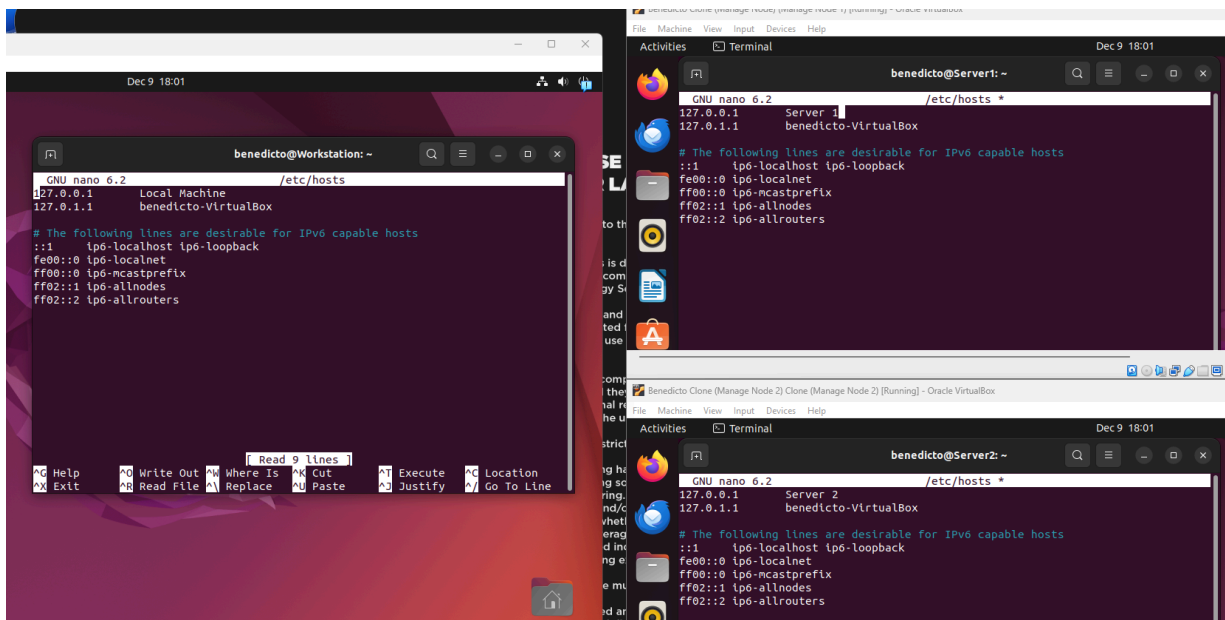
1. Change the hostname using the command *sudo nano /etc/hostname*
 - 1.1 Use server1 for Server 1
 - 1.2 Use server2 for Server 2
 - 1.3 Use workstation for the Local Machine

Screenshot:



2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.
 - 2.1 Type 127.0.0.1 server 1 for Server 1
 - 2.2 Type 127.0.0.1 server 2 for Server 2
 - 2.3 Type 127.0.0.1 workstation for the Local Machine

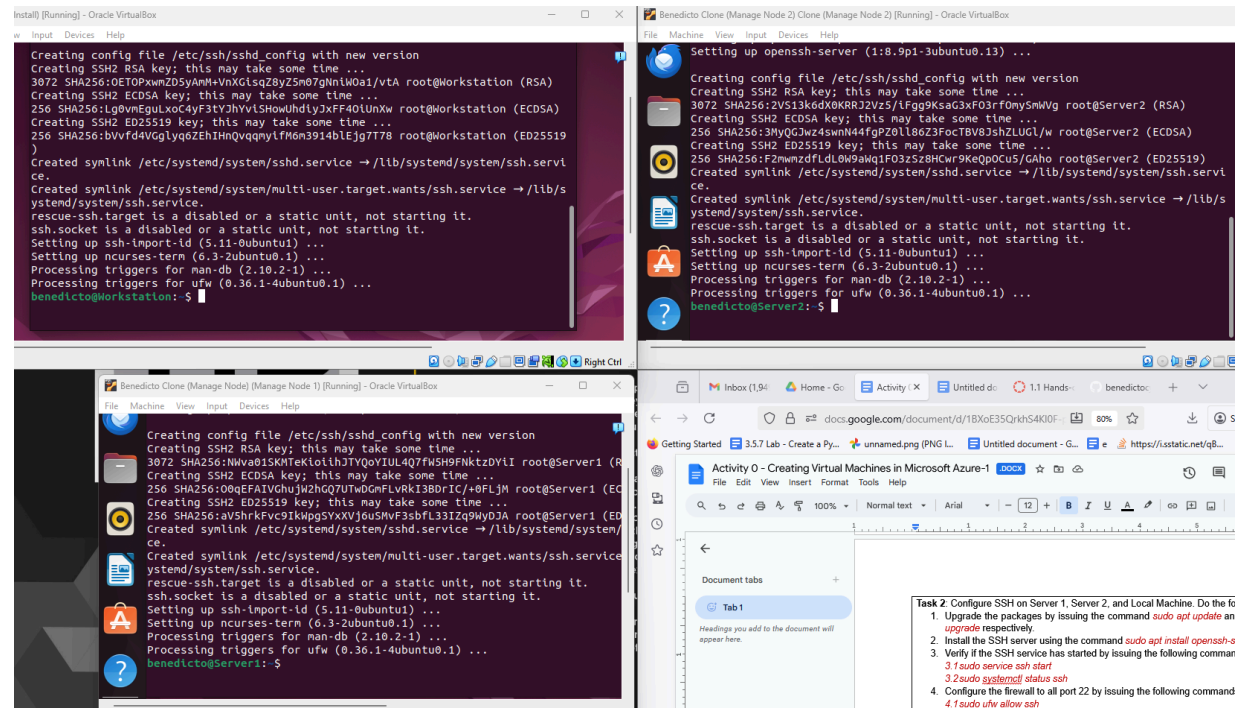
Screenshot:



Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command ***sudo apt update*** and ***sudo apt upgrade*** respectively.
2. Install the SSH server using the command ***sudo apt install openssh-server***.

Screenshot:

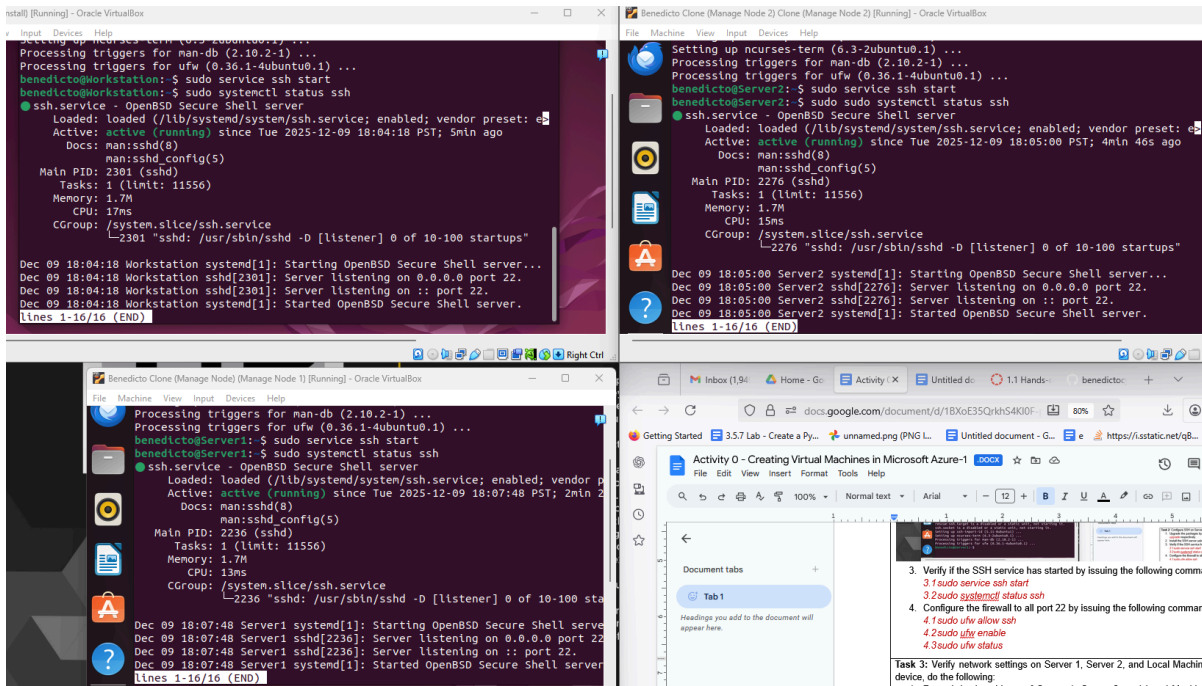


3. Verify if the SSH service has started by issuing the following commands:

3.1 sudo service ssh start

3.2 sudo systemctl status ssh

Screenshot:



4. Configure the firewall to all port 22 by issuing the following commands:

4.1 *sudo ufw allow ssh*

4.2 *sudo ufw enable*

4.3 *sudo ufw status*

Screenshot:

The image contains two screenshots of terminal windows. The left screenshot shows a terminal window titled 'benedicto@Workstation: ~' with a file explorer view on the left showing a directory tree with files like 'NetworkManager.service', 'accounts-daemon.service', 'anacron.service', and 'avahi-daemon.service'. The terminal output shows the following commands and their results:

```
[1]+ Stopped sudo systemctl status
benedicto@Workstation:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
benedicto@Workstation:~$ sudo ufw enable
Firewall is active and enabled on system startup
benedicto@Workstation:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

benedicto@Workstation:~$
```

The right screenshot shows two terminal windows. The top window is titled 'benedicto@Server2: ~' and shows the following commands and their results:

```
[1]+ Stopped sudo systemctl status
benedicto@Server2:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
benedicto@Server2:~$ sudo ufw enable
Firewall is active and enabled on system startup
benedicto@Server2:~$ sudo ufw status
sudo: ufw: command not found
benedicto@Server2:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

benedicto@Server2:~$
```

The bottom window is titled 'benedicto@Server1: ~' and shows the following commands and their results:

```
benedicto@Server1:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
benedicto@Server1:~$ sudo ufw status
Status: inactive
benedicto@Server1:~$ sudo ufw enable
Firewall is active and enabled on system startup
benedicto@Server1:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)

benedicto@Server1:~$
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.

1.1 Server 1 IP address: 192.168.56.104

1.2 Server 2 IP address: 192.168.56.103

1.3 Server 3 IP address: 192.168.56.105

Screenshot:

The image contains two screenshots of terminal windows. The left screenshot shows the configuration of the 'enp0s3' interface on a machine named 'Benedicto@Workstation'. It displays the IP address 192.168.56.103, netmask 255.255.255.0, and broadcast address 192.168.56.255. The right screenshot shows the configuration of the 'enp0s8' interface on a machine named 'Benedicto Clone 1 (Clone 1 (Server 1))'. It displays the IP address 192.168.56.104, netmask 255.255.255.0, and broadcast address 192.168.56.255. Both screenshots also show the status of the interfaces, including RX and TX packets, bytes, and errors.

```
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Processing triggers for man-db (2.12.0-4build2) ...
Benedicto@Workstation:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe8a:afa2 prefixlen 64 scopeid 0x20<link>
    inet6 fd00::73dd:388b:2e98:6ad6 prefixlen 64 scopeid 0x0<global>
    inet6 fd00::a00:27ff:fe8a:afa2 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:8a:af:a2 txqueuelen 1000 (Ethernet)
    RX packets 318 bytes 233307 (233.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 265 bytes 27927 (27.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::f3a9:7a11:d2b5:9238 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:c2:f9:7c txqueuelen 1000 (Ethernet)
    RX packets 25 bytes 5022 (5.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 59 bytes 7549 (7.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

```
enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::9b84:c816:2bae:33ac prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b9:da:6e txqueuelen 1000 (Ethernet)
    RX packets 161 bytes 27316 (27.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 64 bytes 7728 (7.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
```

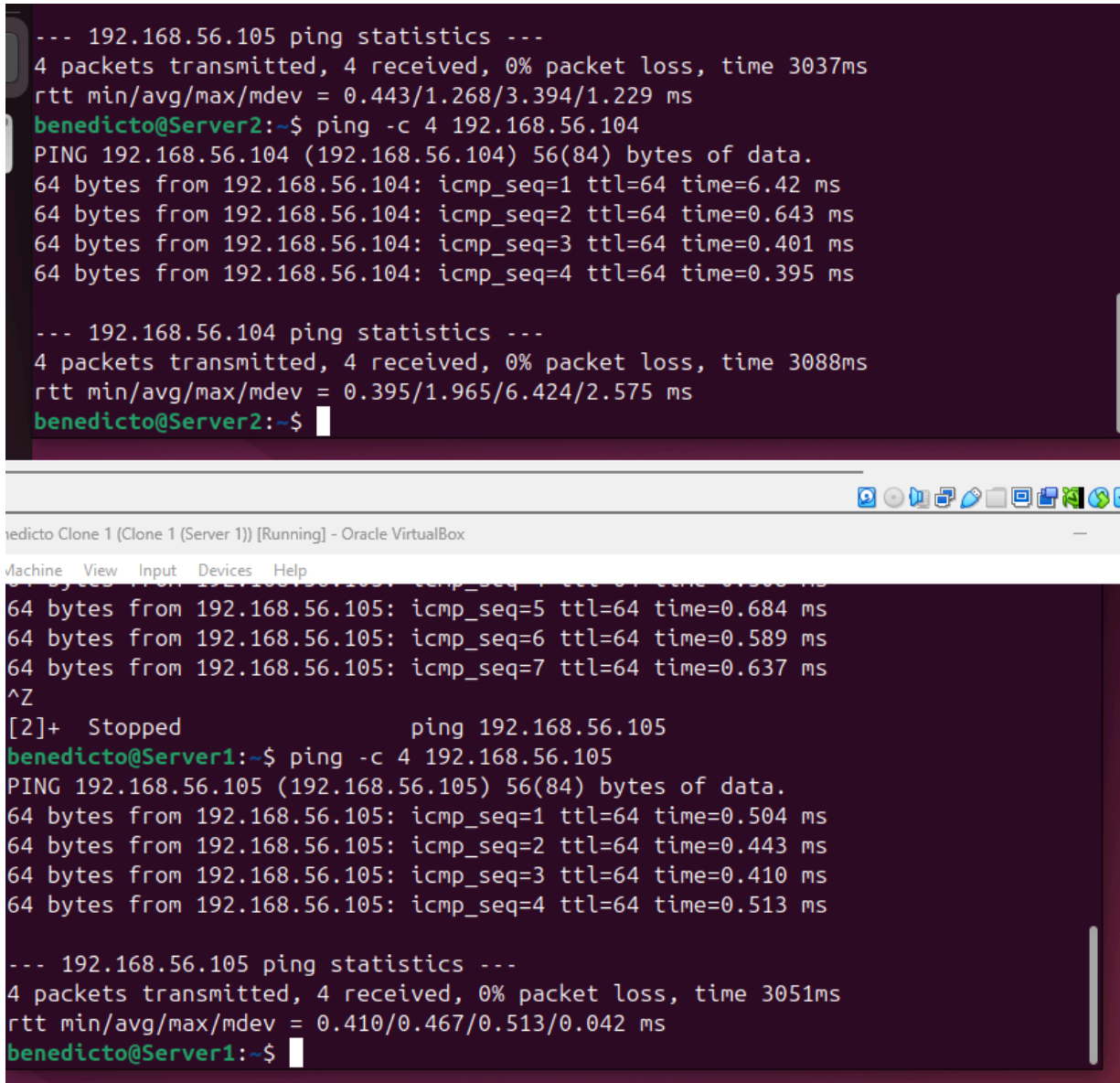
2. Make sure that they can ping each other.

2.1 Connectivity test for Local Machine 1 to Server 1: ☒ Successful ☐ Not Successful

2.2 Connectivity test for Local Machine 1 to Server 2: ☒ Successful ☐ Not Successful

2.3 Connectivity test for Server 1 to Server 2: ☒ Successful ☐ Not Successful

Screenshot:



```
--- 192.168.56.105 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3037ms
rtt min/avg/max/mdev = 0.443/1.268/3.394/1.229 ms
benedicto@Server2:~$ ping -c 4 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=6.42 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=0.643 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=0.401 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=0.395 ms

--- 192.168.56.104 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3088ms
rtt min/avg/max/mdev = 0.395/1.965/6.424/2.575 ms
benedicto@Server2:~$
```

Machine View Input Devices Help

```
64 bytes from 192.168.56.105: icmp_seq=5 ttl=64 time=0.684 ms
64 bytes from 192.168.56.105: icmp_seq=6 ttl=64 time=0.589 ms
64 bytes from 192.168.56.105: icmp_seq=7 ttl=64 time=0.637 ms
^Z
[2]+  Stopped                  ping 192.168.56.105
benedicto@Server1:~$ ping -c 4 192.168.56.105
PING 192.168.56.105 (192.168.56.105) 56(84) bytes of data.
64 bytes from 192.168.56.105: icmp_seq=1 ttl=64 time=0.504 ms
64 bytes from 192.168.56.105: icmp_seq=2 ttl=64 time=0.443 ms
64 bytes from 192.168.56.105: icmp_seq=3 ttl=64 time=0.410 ms
64 bytes from 192.168.56.105: icmp_seq=4 ttl=64 time=0.513 ms

--- 192.168.56.105 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.410/0.467/0.513/0.042 ms
benedicto@Server1:~$
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

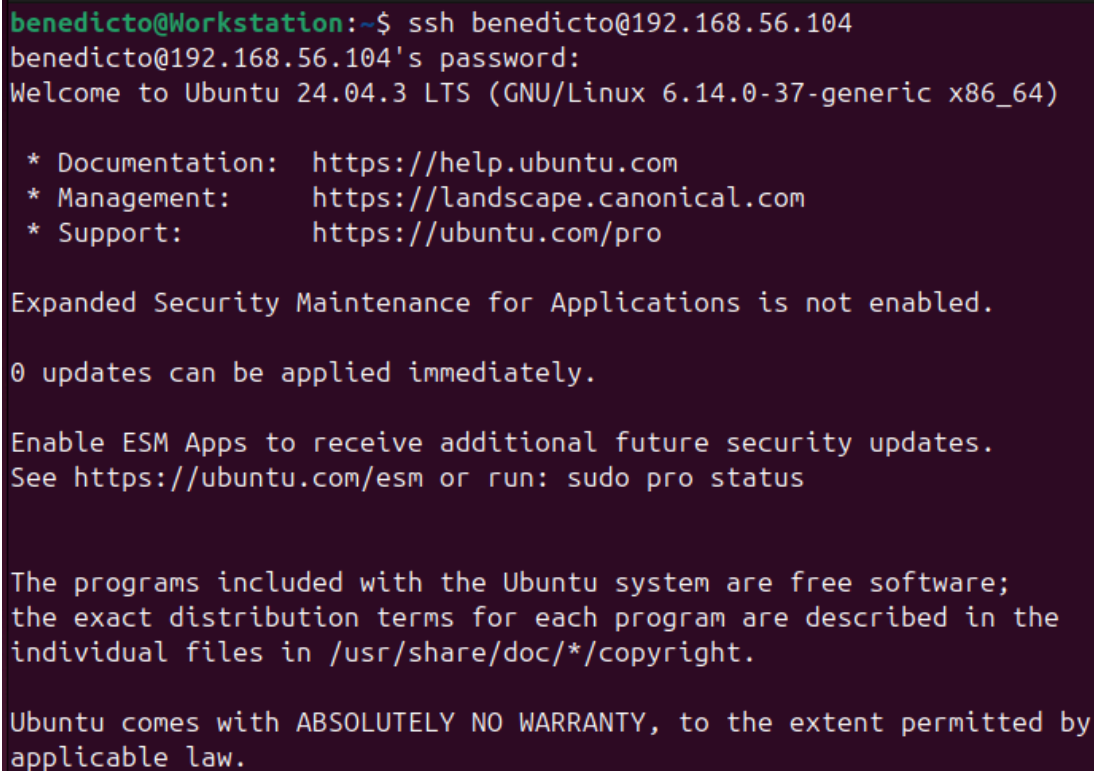
1. On the Local Machine, issue the following commands:

1.1 `ssh username@ip_address_server1` for example, `ssh jvtaylor@192.168.56.120`

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are on server 1. The user should be in this format `user@server1`. For example, `jvtaylor@server1`

Screenshot:



```
benedicto@Workstation:~$ ssh benedicto@192.168.56.104
benedicto@192.168.56.104's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

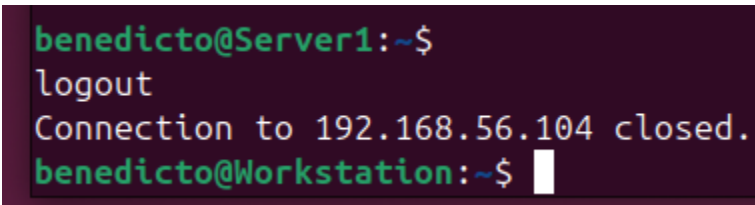
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

2. Logout of Server 1 by issuing the command `control + D`.

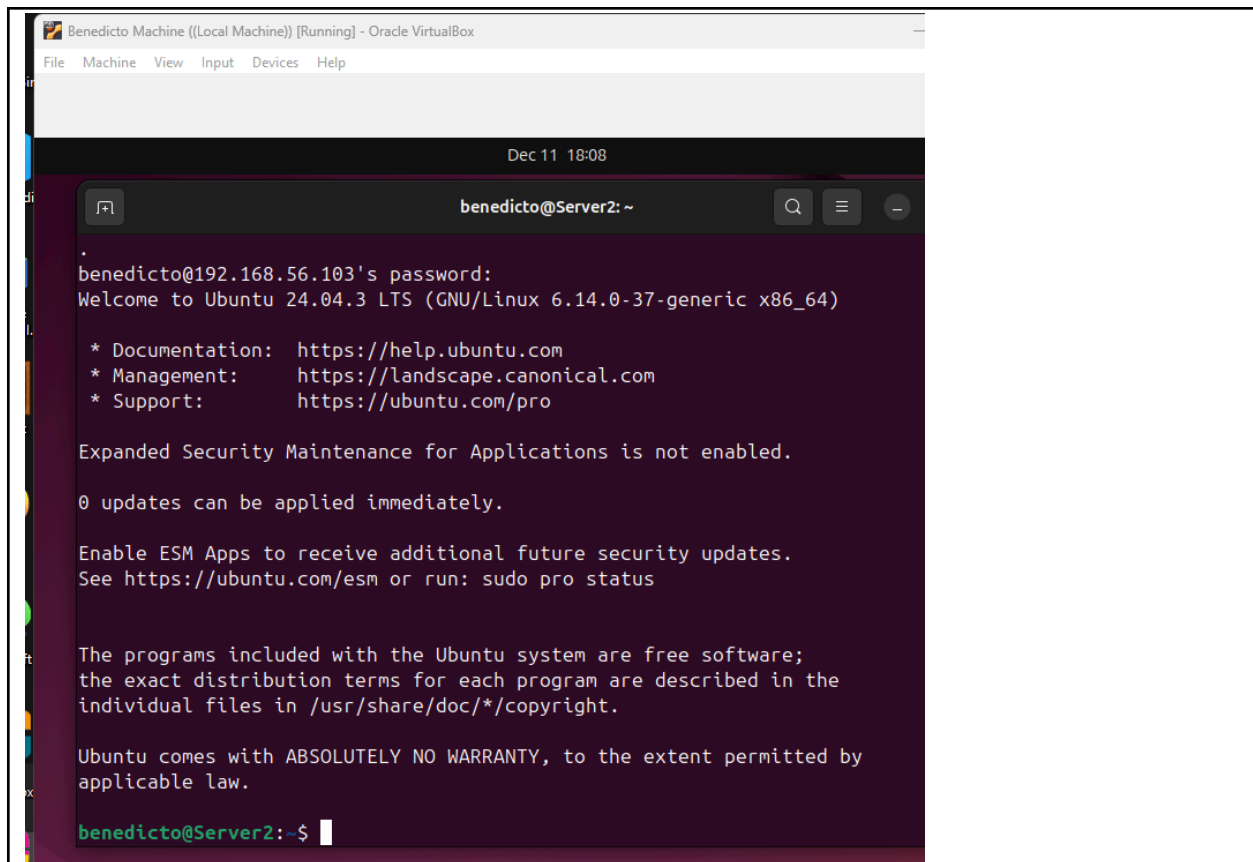
Screenshot:



```
benedicto@Server1:~$
logout
Connection to 192.168.56.104 closed.
benedicto@Workstation:~$
```

3. Do the same for Server 2

Screenshot:



```
Benedicto Machine ((Local Machine)) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Dec 11 18:08
benedicto@Server2: ~
*
benedicto@192.168.56.103's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

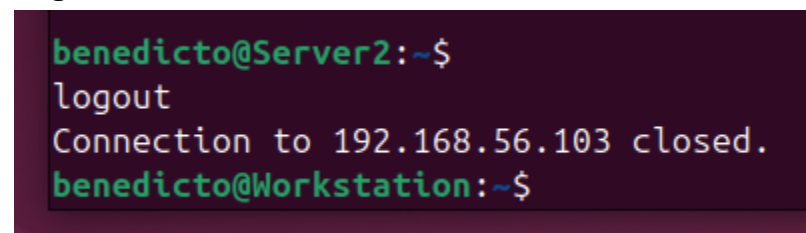
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

benedicto@Server2:~$
```

Logout server 2:



```
benedicto@Server2:~$
logout
Connection to 192.168.56.103 closed.
benedicto@Workstation:~$
```

4. Edit the hosts of the Local Machine by issuing the command *sudo nano /etc/hosts*. Below all texts type the following:
4.1 *IP_address server 1* (provide the ip address of server 1 followed by the hostname)

```
benedicto@server1: ~  
GNU nano 8.3 /etc/hosts *  
192.168.56.103 server1  
127.0.1.1 benedicto-VirtualBox  
  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters
```

4.2 **IP_address server 2** (provide the ip address of server 2 followed by the hostname)

```
benedicto@server2: ~  
GNU nano 8.3 /etc/hosts *  
192.168.56.101 server2  
127.0.1.1 benedicto-VirtualBox  
  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters
```

4.3 Save the file and exit.

5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do **ssh jvtaylor@server1**. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

- **It didn't work.**

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?

- We use hostname by using DNS (Domain Name System) or local hosts files.

2. How secured is SSH?

- Securing SSH involves implementing a layered defense strategy, primarily by replacing password-based logins with more secure methods, restricting user access, and monitoring activity.